

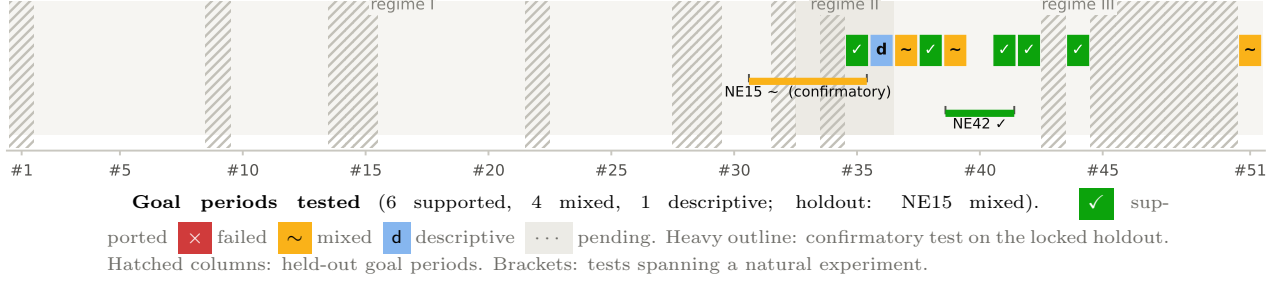
H05: Cutting the cross-room channel lowers entropy production, and rooms become coupled blocks

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-05

Complete 65% Faithfulness 2/5 Usefulness 2.5/5

Bottom line. Rooms partition both chat and artifacts; items leak across mostly through command output and chat relays, and dilution is too small to explain the re-cut jumps.

level: hypothesis · pipeline checks 100% · updated 2026-10-05



WHAT HAPPENED

A kinetic-Ising fit asks whether rooms make agents coupled blocks, on 99 non-holdout days (#35–#44, #51 to 09-04). Within-room talk coupling beats cross-room coupling in 6/6 regime-III two-room windows (Fisher $p = 6 \times 10^{-7}$). The pooled effect of co-location on talk ($z = 5.1$) falls to $z = 0.2$ once ledger read counts enter. **The holdout NE15 split** passes the pair DiD ($\kappa_x - 0.022$, $p_{\text{perm}} = 0.02$) but misses the pooled test, so it is **inconclusive**. **Round 2** asks which channels cross rooms. Rooms split artifacts too: cross-room same-day co-editing is ≤ 0.04 of the within-room rate in 5/8 windows, and NE42 cut pairs go $0.56 \rightarrow 0$ at the split. Cross-room commit coupling through a shared repo is not detectable (power 0.33). Room-size dilution exists ($k^{-0.65}$, post hoc) but predicts only $\times 1.2$ per re-cut, not the $\times 6$. Novel repos leak across rooms through command output at $+0.26$ adoptions per exposure (lift 21); history search shows none.

PHYSICAL MODEL

Model 02 (kinetic Ising). The spin $s_i(t) = \pm 1$ says whether agent i talks (talk spin) or acts (activity spin) in 1-min bin t ; $s'_i = s_i(t+1)$. The field h_i is the daily schedule; a cross-day surrogate removes it. r_i is the room. J_{in} and J_{out} are the same- and cross-room couplings; a cut predicts $J_{\text{out}} \rightarrow 0$. Σ_g is the Aguilera-Ito-Kolchinsky lower bound on the entropy production σ , with dual weights θ . $\hat{R}$ and D are block correlations and variances. α_{ij} and γ_d are pair and day effects.

$$P(s'_i | s) = \frac{e^{s'_i H_i}}{2 \cosh H_i}, \quad H_i = h_i + \sum_j J_{ij} s_j$$

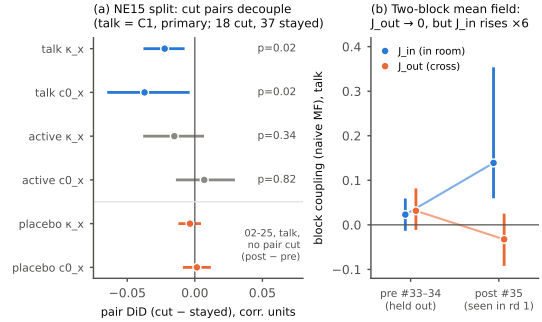
$$J_{ij} = J_{\text{in}} \delta_{r_i r_j} + J_{\text{out}} (1 - \delta_{r_i r_j})$$

$$\sigma = \sum_{i < j} (J_{ij} - J_{ji}) \langle g_{ij} \rangle, \quad g_{ij} = s'_i s_j - s_i s'_j$$

$$\Sigma_g = \max_{\theta} [\theta \cdot \langle g \rangle - \ln \langle e^{-\theta \cdot g} \rangle] \leq \sigma$$

$$J^{\text{MF}} = D^{-1/2} (I - \hat{R}^{-1}) D^{-1/2}$$

$$\kappa_{ij,d} = \alpha_{ij} + \gamma_d + \beta \text{coloc}_{ij,d}$$



Observables. (a) The confirmatory pair difference-in-differences at the 03-16 split: the coupling change of the 18 separated pairs minus that of the 37 pairs that stay together (pre #33–34, held out; post #35, not held out). Talk coupling drops (C1 passes); activity does not. The 02-25 placebo separates no one and shows no change. Bars: day-bootstrap 95% CIs; p : assignment permutation. (b) Block mean field: cross-room J_{out} falls to ≈ 0 , and within-room J_{in} rises (unpredicted).

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

A room is a read filter, and it is also a work filter: the operator's split separates the repositories as well as the chat. To decouple two agents, stop them reading each other. With read counts known, co-location adds nothing ($z = 0.2$). To move one item across rooms, put it in a command's output (a shared listing or README). One such exposure gives about a 1-in-4 adoption within 2 h, $3 \times$ a same-room chat read. History search does not do it. Do not expect a room cut to change pair coupling by attention alone. Dilution gives $\times 1.2$ for a $4 \times$ smaller room; the observed jumps (≈ 0 when merged, $\times 2$ after the split) carry a goal field too.

RESULTS VS PREDICTIONS

Prediction	Result	Verdict
P2 talk within > cross (1b)	+0.021, 6/6, p 6e-7; trim same	yes
P3 split 05-11 DiD < 0 (1b)	-0.020 (p 0.03)	yes
HH248: reads absorb the room (1b)	$\beta(\text{room}) \approx 5.1 \rightarrow 0.2$ given reads	yes
Holdout C1 / C3 / C5	pass / fail / mixed (buggy table)	not re-run
R1-P1 rooms split artifacts	cross/within co-edit ≤ 0.04 in 5/8; NE42 cut pairs $0.56 \rightarrow 0$	yes
R1-P2 cross-room commit coupling	$\Delta -0.017$ (p 0.76); within rooms +0.017 (p 0.015)	inconclusive
R1-P3 #focus keeps commit coupling	DiD -0.086 [-0.135, -0.036]	no
R2-P1/P2 pooled	dilution, $\hat{\beta}_\kappa$ 0.79 (wide); $\hat{\beta}_u$ -0.34 [-0.55, -0.14]	no
R2 two-room era (hoc)	$\hat{\beta}_u$ 0.65 [0.41, 0.90]	near 0.45
R2-P3 NE42 / NE15 sizes	merge drop $>$ dilution; NE15 $\leq 12\%$ of log 6	not dilution
R3-P1 output / leak	+0.26 [0.11, 0.42], lift 21 / 12 exposures	yes / n.p.
R3-P3 leaks < crossings	22% (relay 38%, none logged 41%)	yes

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	0
D unfitted predictions	0	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

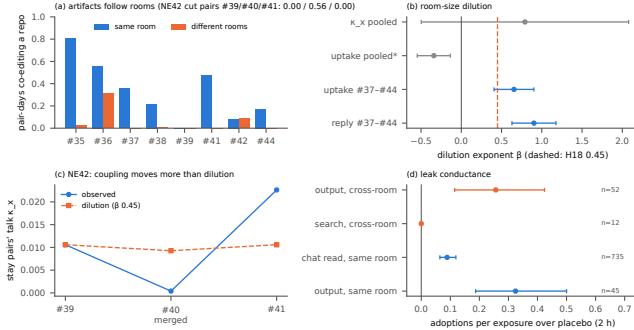
CAVEATS

Co-location and log reads correlate at 0.82, so HH248 has little leverage. Rooms and artifacts are nearly collinear, so the artifact rival is not separable from the room effect. Cross-room commit coupling has power 0.33. The output leak rests on 52 exposures, 48 of them in #36. The pooled dilution test was spoiled by one #focus dyad (no @-mention is needed between two agents). The NE42 size check was not blind. The holdout run used the buggy table.

ROUND 2

Ask Vivian about a fixed-panel NE12/NE15 re-run. Fit a room-specific goal field at NE42 to see whether it absorbs the overshoot. Measure leaks with novel names and numbers (H41 markers), not only repositories.

- **Essence**. Through which channels does information actually move?
- **H05-R1** Rooms matter only insofar as they also split artifacts: cross-room pairs sharing a repo stay coupled through edits (E3).
- **H05-R2** The $\times 6$ rise in within-room coupling after the split is attention reallocation, predicted in size by $N_{\text{room}}^{-0.6}$ (E5).
- **H05-R3** Information leaks across rooms through history search and artifacts at a measurable rate (a leak conductance).



Second figure. Round 2, non-holdout. (a) Share of pair-days on which both agents commit to a common repository, same vs different rooms, per two-room window. (b) Room-size dilution exponents with the room size set by the operator (pair and day FE); dashed: H18's 0.45; grey = pre-registered pooled tests, blue = two-room era (post hoc). (c) NE42 stay pairs' talk coupling vs the dilution prediction. (d) Excess adoptions of a novel repository within 2 h of a first exposure, over matched placebo items of the same agent at the same moment; bars: item bootstrap 95%.

GOAL PERIODS IN DETAIL

NE15 $\sim$ 1b: mixed (not re-run; corrected non-holdout analogue... (**holdout**))

G35 $\checkmark$ 1b: supported p 0.004; fixed bins)

G36 d 1b: descriptive p 0.12; regime II)

G37 $\sim$ 1b: mixed p 0.11)

G38 $\checkmark$ 1b: supported p 0.007; trim +0.018)

G39 $\sim$ 1b: mixed +0.000; was failed)

G41 $\checkmark$ 1b: supported p 0.001)

G42 $\checkmark$ 1b: supported p 0.017; was mixed)

G44 $\checkmark$ 1b: supported p 0.001)

G51 $\sim$ 1b: mixed (the ledger confirms the cut; talk DiD righ...)

+1 more periods: see the card.

H05-rooms-cut generated 2026-10-04 20:39 by infra/summaries/build_summaries.py card hypotheses/H05-rooms-cut/README.md ratings: H05 round-2 agent (suggested); coordinator calibration pending (page 2)